LAB 05

Interpreting Measured Data

Mean, variability, measurement errors and confidence intervals

```
Run time = 0.774 ± 0.066 s (95% CI, 30 runs)
```

System Configuration and Performance Evaluation Lab · Course No: 0613-310

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Same Program, Five Different Times

Lab 02: Sieve of Eratosthenes, N = 10,000,000, idle computer, 5 runs



The slowest run took 29% longer than the fastest. So what is the run time?

Today’s Objectives

1. Summarize many runs with the mean and the median
2. Measure spread with variance, standard deviation, CV and a box plot
3. Tell random errors from systematic errors
4. Compute and explain a 95% confidence interval
5. Work out how many runs a measurement needs

→ Deliverable: Lab Report 05

Why Do Run Times Vary?

Background tasks

Updates, antivirus and the OS wake up and take CPU time

OS scheduling

Our program waits for its turn on the CPU (Lab 02)

Cache and memory

Data may or may not already be in the fast cache

CPU speed changes

Turbo boost, heat and battery saver change the clock speed

Timer limits

Every clock has a smallest step and a small cost to read

The lesson

Measure many times and summarize with statistics

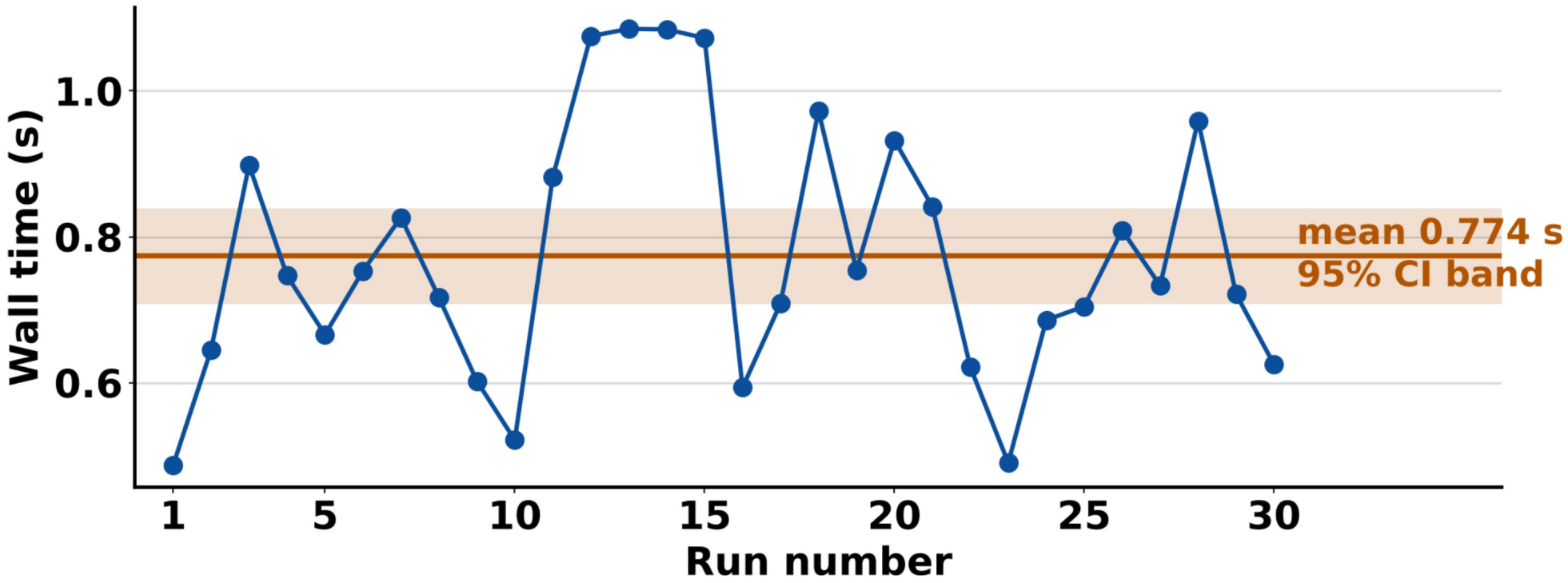
01

Describing the Data

Central tendency · Variability · Box plots

Today's Data: 30 Runs of the Sieve

Sieve of Eratosthenes, N = 5,000,000, idle system, measured with lab5_statistics.py



Central Tendency: One Typical Value

```
Lab 02 runs, sorted: 0.763 0.802 0.835 0.903 0.983 s
```

Mean

Add all values, divide by how many

Σx ÷ n

4.286 ÷ 5 = 0.857 s

Median

Sort the values, take the middle one

middle of sorted list

3rd of 5 values = 0.835 s

Mode

The value that appears most often

most frequent value

No repeats here: no mode

Even number of values? Median = average of the two middle values

Example: One Slow Run

An antivirus scan starts during a 6th run, which takes 2.50 s. What happens?

	5 runs	+ slow run	Change
Mean	**0.857 s**	**1.131 s**	**+32%**
Median	**0.835 s**	**0.869 s**	**+4%**

Mean: one bad value pulls it a long way

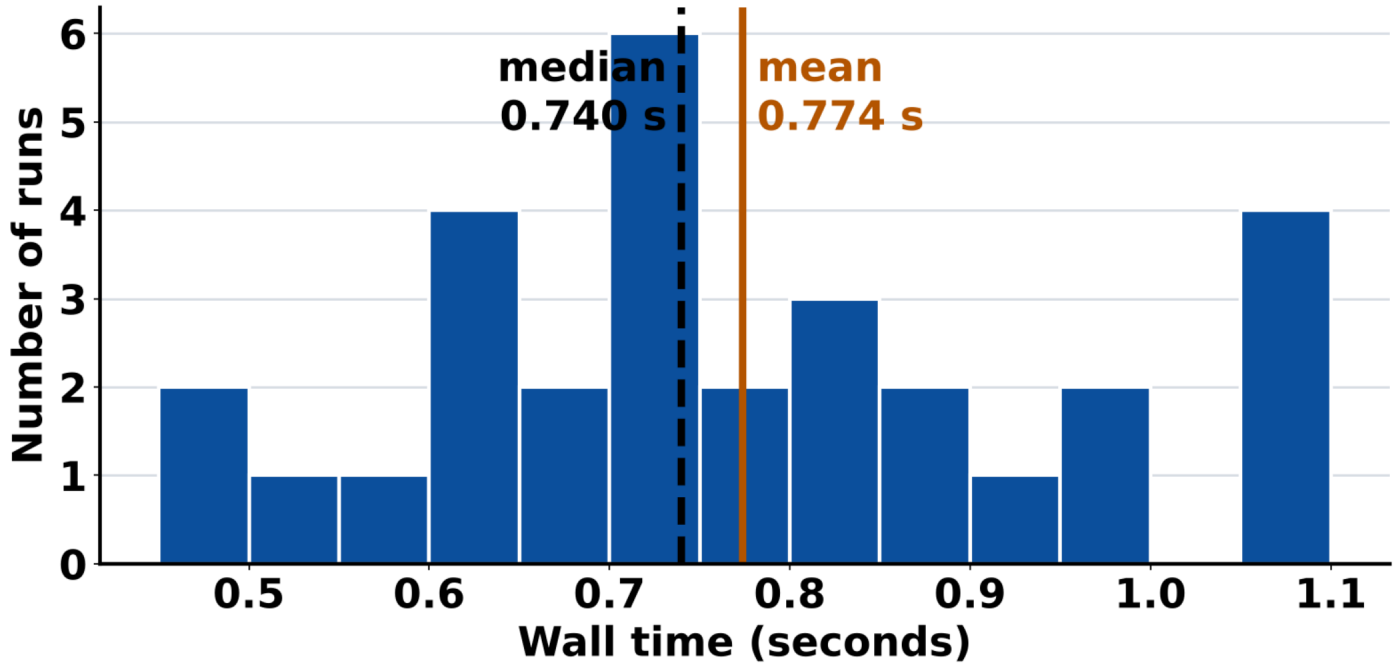
Median: barely moves. It is robust to outliers

Which Summary Should You Use?

Use	When	Example
Mean	Data is roughly symmetric, or the total matters	Total time of 1,000 jobs = 1,000 × mean
Median	Data is skewed or has outliers	Typical run time with a few slow runs
Mode	Data is in categories	Most common error code in a log
Geometric mean	Averaging ratios or speedups	Average speedup over many benchmarks (SPEC)

> **When in doubt, report both the mean and the median**

Histogram: The Shape of the Data



Mean > median

0.774 s vs 0.740 s: a few slow runs pull the mean right

Why times skew right

A run can be slowed down a lot, but never run faster than the hardware allows

Variability: How Spread Out?

Range

`max - min`

Simple, but set by the two most extreme runs only

Variance (s^2)

$\Sigma(x - \text{mean})^2 \div (n - 1)$

Average squared distance from the mean; unit is s^2

Standard deviation (s)

$\sqrt{\text{variance}}$

Typical distance from the mean, in seconds

Coefficient of variation (CV)

$s \div \text{mean} \times 100\%$

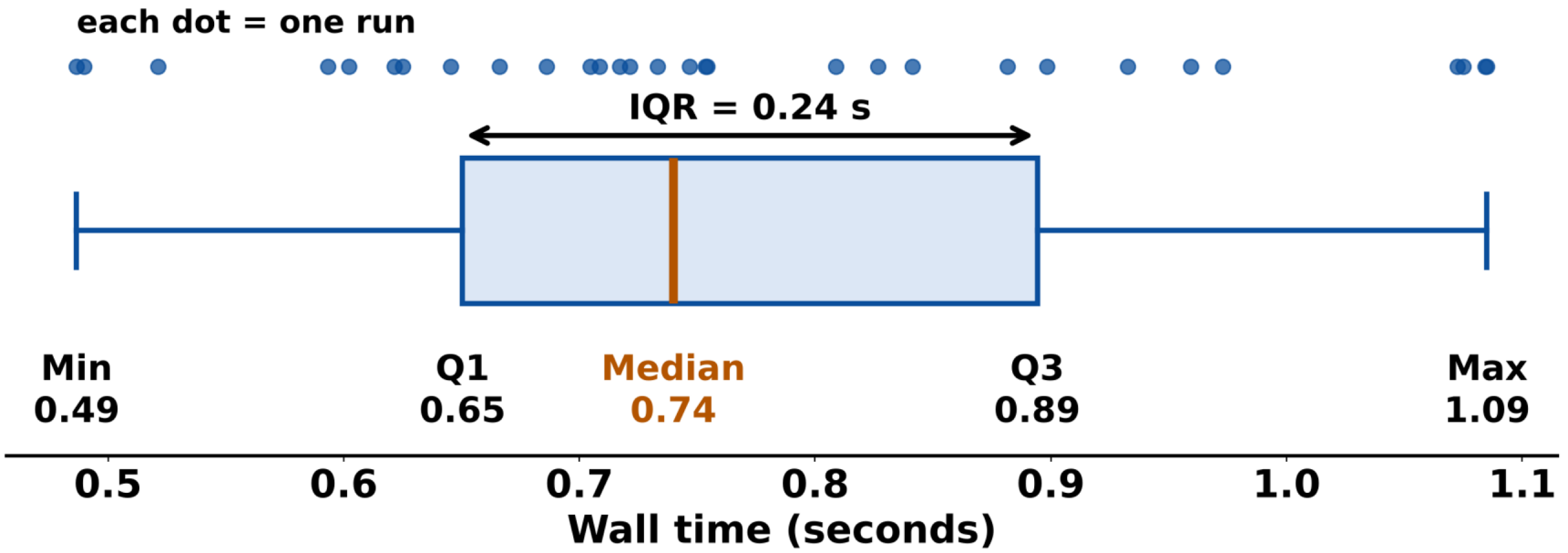
Spread as a percentage: compares different programs

Worked Example: Standard Deviation

Run	x (s)	x − mean	(x − mean)²
1	0.9831	+0.1259	0.015846
2	0.8015	−0.0557	0.003105
3	0.8352	−0.0220	0.000485
4	0.9032	+0.0460	0.002114
5	0.7631	−0.0941	0.008859
Sum	4.2861	0	0.030408

1. mean = 4.2861 ÷ 5 = 0.857 s
2. variance = 0.030408 ÷ 4 = 0.0076 s^2
3. s = √0.0076 = 0.087 s
4. CV = 0.087 ÷ 0.857 = 10.2%
5. range = 0.983 − 0.763 = 0.220 s

Quartiles and the Box Plot



Box = middle 50% of runs (Q1 to Q3)

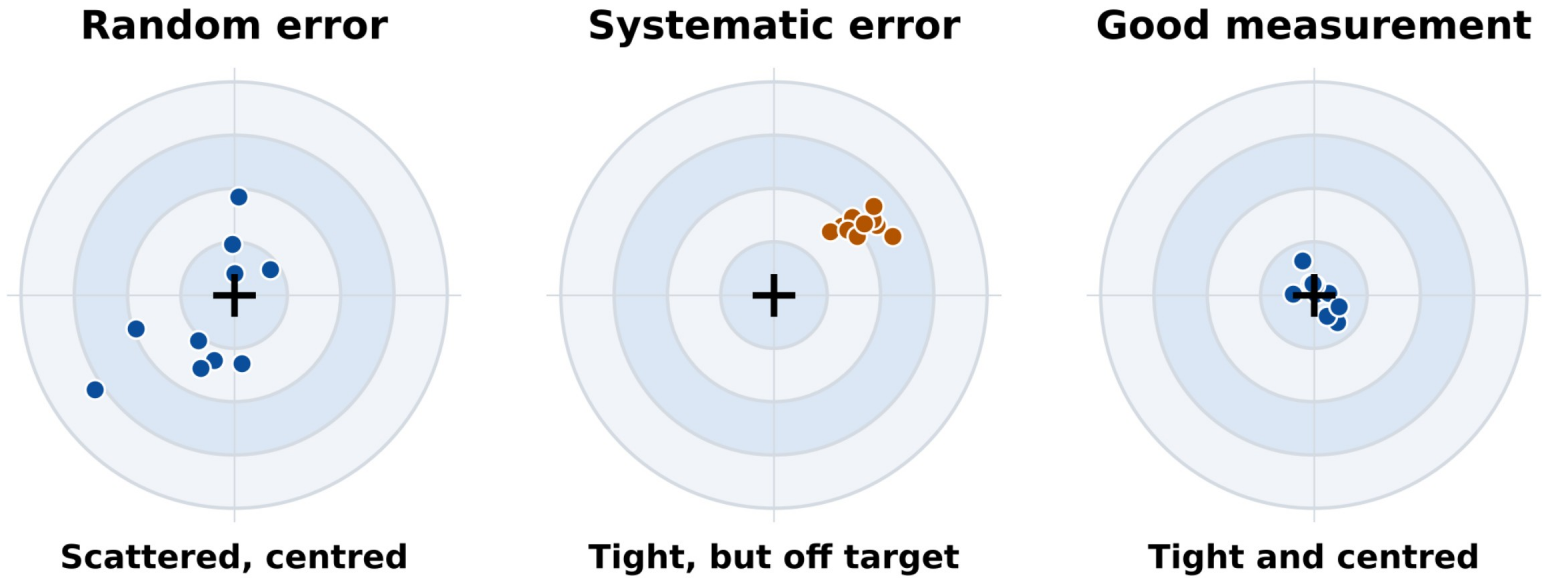
Outlier rule: below Q1 − 1.5 × IQR or above Q3 + 1.5 × IQR

02

Measurement Errors

Random errors · Systematic errors · How to reduce them

Two Kinds of Error



Random error

Changes every run, up or down. More runs average it out.

Systematic error

Same direction every run. More runs do not fix it.

Errors in Our Own Labs

Error	Type	Fix
First run slower: code and data not yet in cache (warm-up)	Systematic	Add a warm-up run, or drop run 1
Antivirus or update starts during a run	Random	More runs; report the median
Battery saver lowers the CPU speed	Systematic	Plug in; same power mode
Profiler running while timing (Lab 04)	Systematic	Time with the profiler off
Job too short for the timer	Systematic	perf_counter(); runs of 0.1 s or more

The program checks for warm-up: it compares run 1 with the rest

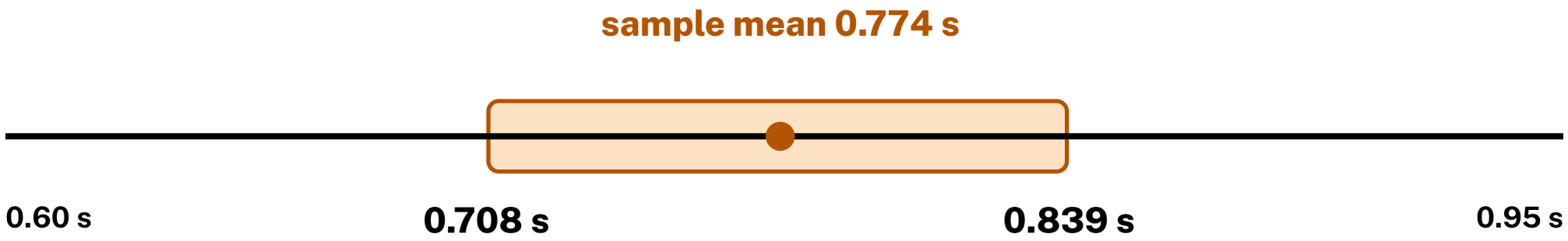
03

Confidence Intervals

How sure are we? · How many runs do we need?

The Idea: A Range for the True Mean

Our 30 runs are only a sample. The true mean (μ), the average over endless runs, is unknown.



> **“We are 95% confident the true mean run time is between 0.708 s and 0.839 s.”**

The Formula

$$CI = \bar{x} \pm t \times s \div \sqrt{n}$$

$\bar{x}$
sample mean

s
standard deviation

n
number of runs

t
from the t-table, n − 1 degrees of freedom

Runs n	t (95%)
5	2.776
10	2.262
20	2.093
30	2.045
very many	1.960

$t \times s \div \sqrt{n}$ is the margin of error: half the width of the interval

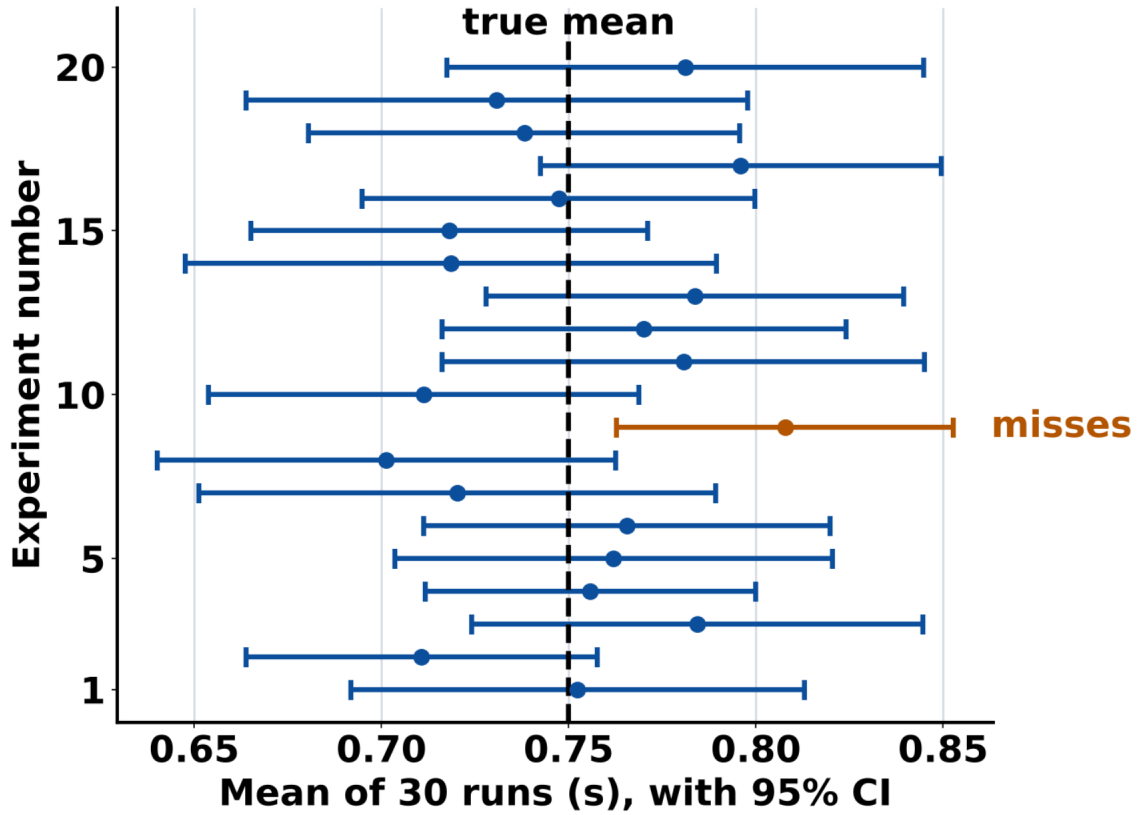
Worked Example: 95% CI for 5 Runs

Lab 02 runs: mean = 0.857 s, s = 0.0872 s, n = 5

1	Look up t	`n - 1 = 4 → t = 2.776`
2	Standard error s ÷ √n	`0.0872 ÷ 2.236 = 0.0390 s`
3	Margin of error	`2.776 × 0.0390 = 0.108 s`
4	Interval	`0.857 ± 0.108 = [0.749, 0.965] s`

±0.108 s is ±12.6% of the mean: 5 runs give only a rough answer

What Does “95%” Really Mean?



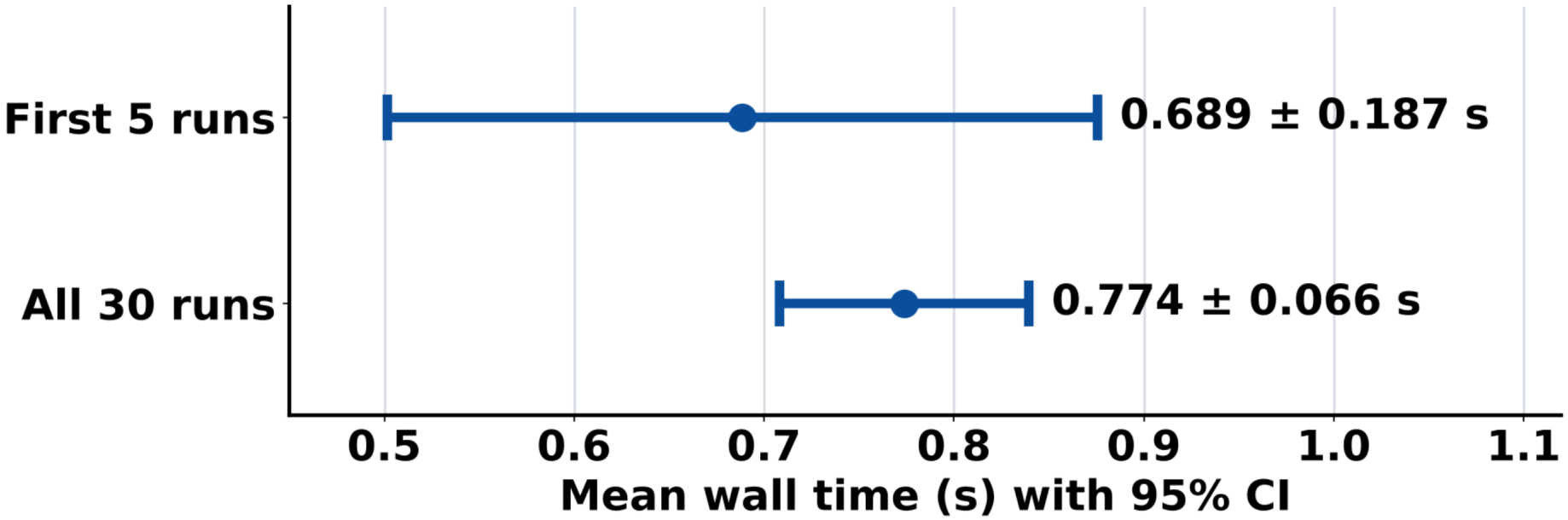
Correct

Repeat the whole experiment 20 times: about 19 of the 20 intervals contain the true mean

Wrong

“95% of the runs fall inside the interval.” The CI is about the mean, not single runs

More Runs, Narrower Interval



Same experiment: 5 runs → ±0.187 s; 30 runs → ±0.066 s

Margin shrinks with √n: 4× the runs → half the width

How Many Runs Do We Need?

$$n = (100 \times z \times s \div (r \times \text{mean}))^2$$

z = 1.96 for 95% · s = SD from a few trial runs · r = precision you want, in % (±r% of the mean)

Data	CV	For ±5%	For ±1%
Lab 02 sample (5 runs)	10.2%	16 runs	398 runs
Today's test (30 runs)	22.7%	80 runs	1,984 runs

A noisy system needs far more runs: quieten the system before adding runs

04

The Program and the Lab

lab5_statistics.py · lab5_compare_ci.py · Lab Report 05

Code 1: Run 30 Times and Record

```
N = 5_000_000          # primes up to N
RUNS = 30              # number of measurements
CONDITION = "normal"   # change to "loaded" under load

def measure_runs(runs):
    times = []
    for run in range(1, runs + 1):
        start = time.perf_counter()
        count = sieve(N)
        times.append(time.perf_counter() - start)
    return times
```

Every run is kept: all statistics come from this list of 30 times

Code 2: Describe the Data

```
data = np.array(times)
mean     = data.mean()
median   = np.median(data)
variance = data.var(ddof=1)     # divide by n - 1
sd       = data.std(ddof=1)
cv       = sd / mean * 100

q1, q3 = np.percentile(data, [25, 75])
iqr = q3 - q1
low_fence, high_fence = q1 - 1.5 * iqr, q3 + 1.5 * iqr
outliers = [t for t in times if t < low_fence or t > high_fence]
```

> **ddof=1 is essential: NumPy divides by n unless you tell it otherwise**

Code 3: Confidence Interval and Runs Needed

```
def confidence_interval(data, confidence=0.95):
    n = len(data)
    mean = statistics.mean(data)
    sd = statistics.stdev(data)
    t_value = stats.t.ppf((1 + confidence) / 2, df=n - 1)
    half_width = t_value * sd / math.sqrt(n)
    return mean, half_width, mean - half_width, mean + half_width

def runs_needed(data, precision_percent, z=1.96):
    mean, sd = statistics.mean(data), statistics.stdev(data)
    return math.ceil((100 * z * sd / (precision_percent * mean)) ** 2)
```

stats.t.ppf(0.975, 4) = 2.776, the same value as the t-table

Sample Output: Summary of 30 Runs

```
  Mean                 = 0.7738 s
  Median               = 0.7399 s
  Range                = 0.5984 s
  Variance             = 0.030913 s^2
  Standard deviation   = 0.1758 s
  Coeff. of variation  = 22.72 %
  IQR                  = 0.2438 s
  Outliers             : none
  First run (warm-up)  = 0.4867 s   vs mean of runs 2-30: 0.7837 s

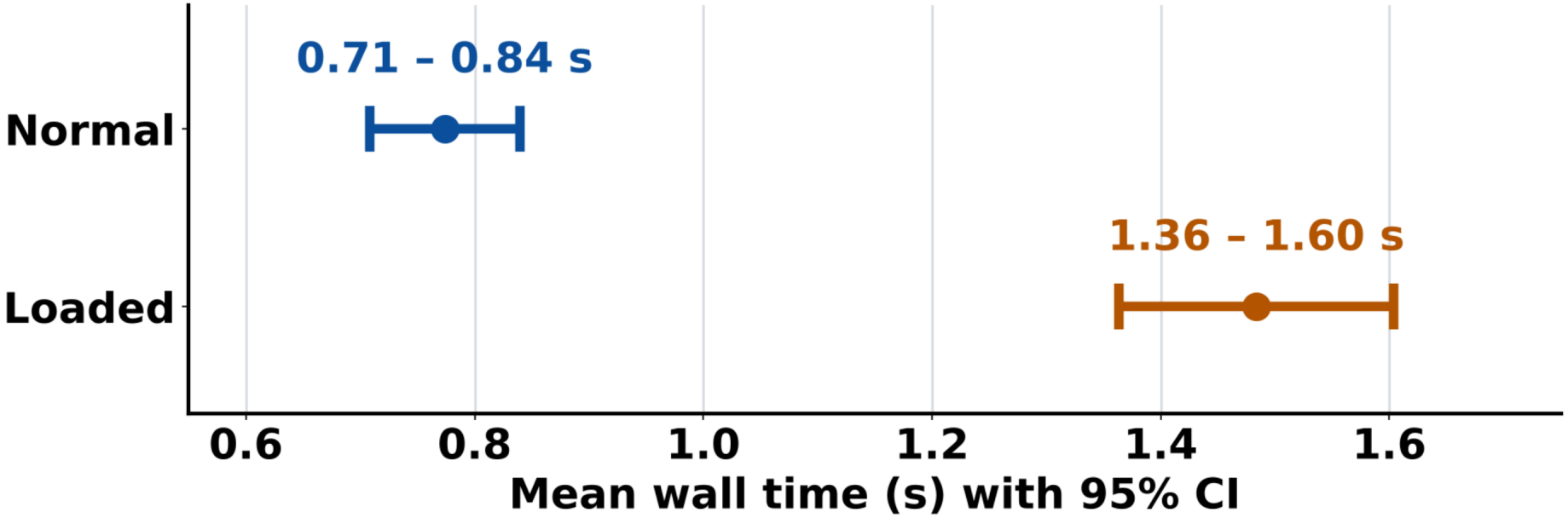
95% confidence interval of the mean
  First 5 runs            : 0.6886 +/- 0.1868 s  →  [0.5018, 0.8754]  (+/- 27.1 %)
  All 30 runs             : 0.7738 +/- 0.0657 s  →  [0.7081, 0.8394]  (+/- 8.5 %)
  Runs 2-30 (no warm-up)  : 0.7837 +/- 0.0647 s  →  [0.7189, 0.8484]  (+/- 8.3 %)

Runs needed for a given precision (z = 1.96)
  +/- 5% of the mean   : 80 runs
  +/- 1% of the mean   : 1984 runs
```

Lab Task

Setup	`pip install numpy scipy matplotlib`
Step A	Idle system, CONDITION = "normal": run python lab5_statistics.py; save the output and the charts
Step B	Set CONDITION = "loaded", start the busy loops from Lab 02, and run it again
Step C	Stop the loads, then run python lab5_compare_ci.py to compare the two intervals
Report	Fill the results table, attach both charts, answer the six questions

Example: Normal vs Loaded



No overlap → the difference is real (statistically significant)

Overlap → not sure yet: use a t-test (Lab 06)

Results Table for the Report

Statistic	Normal	Loaded
Mean (s)		
Median (s)		
Standard deviation (s)		
Coefficient of variation (%)		
Minimum / maximum (s)		
IQR (s) and outliers		
First run vs mean of the rest (s)		
95% CI, first 5 runs (s)		
95% CI, all 30 runs (s)		
Runs needed for ±5%		

Analysis Questions

1. Is the mean or the median the better summary of your runs? Use your histogram.
2. Compare the 95% CI from the first 5 runs with the CI from all 30. Why is one narrower?
3. Was run 1 different from the rest? Is that a random or a systematic error?
4. Did you find any outliers? What might have caused them?
5. How many runs would ±1% precision need? Is that practical?
6. Do the Normal and Loaded intervals overlap? What does that tell you?

> **Answer with your own numbers, and report the CI in a full sentence**

Common Mistakes to Avoid

Mistake	Fix
np.std(data) without ddof=1	Sample SD needs n − 1: use ddof=1
Using z = 1.96 with only 5 runs	Use t: 2.776 for 5 runs
“95% of runs are inside the CI”	The CI is a range for the mean
Deleting slow runs because they look bad	Remove a run only with a reason, and say so
Reporting only the mean	Mean (or median) + SD + 95% CI
Changing N or the load halfway through	Same settings for all 30 runs

Key Takeaway

One measurement is never enough. Report a typical value, its spread, and how sure you are.

```
0.774 ± 0.066 s (95% CI, 30 runs, CV 22.7%)
```

Next lab: Comparing alternatives with the t-test and ANOVA